EA-0026 - AQELYN Engineering Archive

AQELYN - EA-0026 Engineering Archive

IS-026 - AQELYN Configuration Compliance & Drift Intelligence Engine

Archive ID: EA-0026

Implementation Specification: IS-026

Component: AQELYN Configuration Compliance & Drift Intelligence Engine

Project: AQELYN

System Type: Cyber Security Operating Environment

Status: COMPLETE

Repository Impact: No top-level repository structure changes

Breaking Changes: None

Predecessor Archives: EA-0001 through EA-0025

Next Specification: IS-027 - AQELYN Identity Threat Detection & Behavioral Analytics Engine

Document Control

Field	Value
Document	Engineering Archive EA-0026
Specification	IS-026 - AQELYN Configuration Compliance & Drift Intelligence Engine
Publication Format	Markdown, PDF, HTML, ZIP
Source of Truth	MD/EA-0026.md
Archive Rule	Implementation Specification -> Engineering Archive -> Continue
Repository Rule	Fixed repository structure; no redesign
Completion State	IS-026 complete; EA-0026 generated

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

Engineering Archive	Implementation Specification	Component
EA-0001	IS-001	AQELYN Kernel
EA-0002	IS-002	Universal Object Model
EA-0003	IS-003	AQELYN Event Bus
EA-0004	IS-004	AQELYN Evidence Engine

| EA-0005 | IS-005 | AQELYN Knowledge Graph |

| EA-0006 | IS-006 | AQELYN Trust Engine |

| EA-0007 | IS-007 | AQELYN Mission Engine |

| EA-0008 | IS-008 | AQELYN Workflow Engine |

| EA-0009 | IS-009 | AQELYN Policy Engine |

| EA-0010 | IS-010 | AQELYN Compliance & Governance Engine |

| EA-0011 | IS-011 | AQELYN Identity & Access Governance Engine |

| EA-0012 | IS-012 | AQELYN Asset & Configuration Governance Engine |

| EA-0013 | IS-013 | AQELYN Risk Intelligence Engine |

| EA-0014 | IS-014 | AQELYN Threat Intelligence Fusion Engine |

| EA-0015 | IS-015 | AQELYN Security Operations Engine |

| EA-0016 | IS-016 | AQELYN Digital Forensics Engine |

| EA-0017 | IS-017 | AQELYN Threat Detection & Analytics Engine |

| EA-0018 | IS-018 | AQELYN Automated Response & Orchestration Engine |

| EA-0019 | IS-019 | AQELYN Security Data Lake & Telemetry Platform |

| EA-0020 | IS-020 | AQELYN AI Decision Intelligence Engine |

| EA-0021 | IS-021 | AQELYN Predictive Analytics & Forecasting Engine |

| EA-0022 | IS-022 | AQELYN Executive Intelligence & Strategic Reporting Engine |

| EA-0023 | IS-023 | AQELYN Threat Exposure & Attack Surface Management Engine |

| EA-0024 | IS-024 | AQELYN Vulnerability Intelligence & Prioritization Engine |

| EA-0025 | IS-025 | AQELYN Cyber Asset Discovery & Inventory Intelligence Engine |

| EA-0026 | IS-026 | AQELYN Configuration Compliance & Drift Intelligence Engine |

The fixed repository structure remains:

```
AQELYN/
■■■ archive/
■■■ blueprint/
■■■ docs/
■■■ src/
■■■ tests/
■■■ tools/
■■■ build/
■■■ releases/
■■■ scripts/
■■■ assets/
■■■ examples/
■■■ plugins/
■■■ sdk/
■■■ api/
■■■ README.md
```

2. IS-026 Specification Identity

```
Specification ID: IS-026
Name: AQELYN Configuration Compliance & Drift Intelligence Engine
Engineering Archive Target: EA-0026
Project: AQELYN
System Type: Cyber Security Operating Environment
Status: Complete
Predecessor: IS-025 - AQELYN Cyber Asset Discovery & Inventory Intelligence Engine
```

3. Purpose

The AQELYN Configuration Compliance & Drift Intelligence Engine provides continuous configuration assessment, baseline enforcement, drift detection, compliance validation, policy correlation, remediation recommendations, and configuration governance across enterprise infrastructure.

The engine establishes configuration integrity by continuously comparing observed configurations against approved baselines, security policies, regulatory requirements, and organizational standards.

It answers:

```
Which assets are configuration compliant?
Which configurations have drifted?
What configuration changes introduce risk?
Which policy violations exist?
Which systems require remediation?
Can configuration history be explained and audited?
```

4. Mission

The engine shall provide:

```
Configuration baseline management
Continuous compliance validation
Configuration drift detection
Policy compliance assessment
Remediation recommendations
Configuration intelligence
Mission-aware compliance reporting
Executive compliance summaries
Continuous reassessment
Configuration governance
```

5. Scope

5.1 In Scope

```
Operating system configurations
Cloud configurations
Network device configurations
Container configurations
Application configurations
Database configurations
Identity security settings
Infrastructure-as-Code configurations
```

5.2 Out of Scope

```
Application source code
Software development lifecycle
Patch deployment
License management
Financial compliance
```

6. Dependencies

IS-026 depends on:

```
IS-001 AQELYN Kernel
IS-002 Universal Object Model
IS-003 AQELYN Event Bus
IS-004 AQELYN Evidence Engine
IS-005 AQELYN Knowledge Graph
IS-006 AQELYN Trust Engine
IS-007 AQELYN Mission Engine
IS-008 AQELYN Workflow Engine
```

```
IS-009 AQELYN Policy Engine
IS-010 AQELYN Compliance & Governance Engine
IS-011 Identity & Access Governance Engine
IS-012 Asset & Configuration Governance Engine
IS-013 Risk Intelligence Engine
IS-014 Threat Intelligence Fusion Engine
IS-015 Security Operations Engine
IS-016 Digital Forensics Engine
IS-017 Threat Detection & Analytics Engine
IS-018 Automated Response & Orchestration Engine
IS-019 Security Data Lake & Telemetry Platform
IS-020 AI Decision Intelligence Engine
IS-021 Predictive Analytics & Forecasting Engine
IS-022 Executive Intelligence & Strategic Reporting Engine
IS-023 Threat Exposure & Attack Surface Management Engine
IS-024 Vulnerability Intelligence & Prioritization Engine
IS-025 Cyber Asset Discovery & Inventory Intelligence Engine
```

7. High-Level Architecture

```
AQELYN Configuration Compliance & Drift Intelligence Engine
■
■■■ Baseline Management Engine
■■■ Compliance Assessment Engine
■■■ Drift Detection Engine
■■■ Configuration Intelligence Engine
■■■ Remediation Recommendation Engine
■■■ Policy Correlation Engine
■■■ Knowledge Graph Connector
■■■ Security Data Lake Connector
■■■ AI Decision Connector
■■■ Executive Reporting Connector
■■■ Event Publisher
```

8. Functional Requirements

FR-026-001 - Configuration Baselines

The engine shall manage approved configuration baselines for:

```
Servers
Endpoints
Cloud resources
Containers
Network devices
Applications
Databases
```

FR-026-002 - Continuous Compliance

Continuously validate:

```
Baseline compliance
Security policies
Regulatory requirements
Configuration integrity
Asset configuration status
```

FR-026-003 - Drift Detection

Detect:

```
Unauthorized changes
Baseline deviations
Policy violations
```

```
Configuration anomalies
Risk-inducing modifications
```

FR-026-004 - Explainable Compliance Intelligence

Every assessment shall include:

```
Evidence references
Confidence indicators
Baseline references
Policy rationale
Configuration history
Drift explanation
```

FR-026-005 - Governance

Support:

```
Approval workflows
Policy validation
Version control
Auditability
Executive review
```

FR-026-006 - Event Publication

Publish standardized events:

```
configuration.assessed
configuration.drift.detected
configuration.compliant
configuration.noncompliant
baseline.updated
remediation.recommended
```

9. Non-Functional Requirements

The engine shall provide:

```
Continuous assessment
Enterprise scalability
Low-latency drift detection
Explainability
Auditability
Repository stability
Backward compatibility
```

10. Core Configuration Lifecycle

```
Configuration Collected
↓
Baseline Comparison
↓
Compliance Assessment
↓
Drift Detection
↓
Remediation Recommendation
↓
Policy Validation
↓
Continuous Monitoring
```

11. Internal Component Architecture

The AQELYN Configuration Compliance & Drift Intelligence Engine is implemented as a modular configuration intelligence platform integrated with the AQELYN Kernel, Knowledge Graph, Security Data Lake, Asset Inventory Engine, Policy Engine, Compliance Engine, Risk Intelligence Engine, AI Decision Intelligence Engine, and Executive Intelligence Engine.

```
AQELYN Configuration Compliance & Drift Intelligence Engine
■
■■■ Baseline Management Engine
■■■ Compliance Assessment Engine
■■■ Drift Detection Engine
■■■ Configuration Intelligence Engine
■■■ Remediation Recommendation Engine
■■■ Policy Correlation Engine
■■■ Knowledge Graph Connector
■■■ Security Data Lake Connector
■■■ AI Decision Connector
■■■ Executive Reporting Connector
■■■ Event Publisher
```

12. Component Specifications

12.1 Baseline Management Engine

Maintains approved enterprise configuration baselines.

Capabilities:

```
Baseline creation
Baseline versioning
Baseline approval
Baseline distribution
Baseline lifecycle management
```

12.2 Compliance Assessment Engine

Continuously validates configuration compliance.

Supports:

```
Baseline comparison
Policy validation
Regulatory assessment
Configuration verification
Compliance scoring
```

12.3 Drift Detection Engine

Detects configuration deviations.

Produces:

```
Configuration drift
Unauthorized changes
Security deviations
Configuration anomalies
Risk-inducing modifications
```

12.4 Configuration Intelligence Engine

Maintains configuration intelligence.

Produces:

```
Configuration history
Configuration health
Configuration risk
Mission impact
Business impact
```

12.5 Remediation Recommendation Engine

Generates remediation guidance.

Supports:

```
Configuration corrections
Policy recommendations
Baseline restoration
Compensating controls
Executive summaries
```

12.6 Policy Correlation Engine

Maps configurations to enterprise policies.

Produces:

```
Policy mappings
Control mappings
Regulatory mappings
Compliance evidence
Governance status
```

13. Universal Object Model Extensions

13.1 ConfigurationRecord

```
ConfigurationRecord:
configuration_id
asset_id
baseline_version
compliance_status
```

13.2 BaselineDefinition

```
BaselineDefinition:
baseline_id
version
approval_status
effective_date
```

13.3 DriftAssessment

```
DriftAssessment:
assessment_id
drift_type
confidence
detected_at
```

13.4 ConfigurationRemediation

```
ConfigurationRemediation:
remediation_id
recommendation
owner
target_date
```

14. Knowledge Graph Integration

Relationships:

```
Asset
↓
has
↓
Configuration

Configuration
↓
validated_against
↓
Baseline

Configuration
↓
violates
↓
Policy

Policy
↓
supports
↓
Compliance
```

15. Event Bus Integration

15.1 Configuration Events

```
configuration.assessed
configuration.updated
configuration.compliant
configuration.noncompliant
```

15.2 Drift Events

```
configuration.drift.detected
configuration.drift.resolved
```

15.3 Baseline Events

```
baseline.created
baseline.updated
baseline.retired
```

15.4 Remediation Events

```
remediation.recommended
remediation.completed
```

16. Security Data Lake Integration

Consumes:

```
Configuration telemetry
Historical configurations
Change history
Asset metadata
Compliance telemetry
```

17. AI Decision Intelligence Integration

Consumes:

```
Compliance scores
Drift analysis
Remediation recommendations
Configuration trends
Learning insights
```

18. Risk & Asset Integration

Consumes:

```
Asset inventory
Risk scores
Mission impact
Business criticality
Vulnerability intelligence
```

19. Compliance Integration

Supports:

```
Regulatory reporting
Policy validation
Control mapping
Audit evidence
Governance reporting
```

20. Public APIs

20.1 Configuration API

```
GET /configurations
POST /configurations
GET /configurations/{id}
```

20.2 Baseline API

```
GET /baselines
POST /baselines
PUT /baselines/{id}
```

20.3 Drift API

```
GET /drift
GET /drift/{id}
```

20.4 Compliance API

```
GET /compliance-summary
GET /configuration-health
```

21. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/
■■■ src/
■ ■■■ configuration_compliance/
■■■ tests/
■ ■■■ configuration_compliance/
■■■ docs/
■ ■■■ configuration_compliance/
■■■ api/
■ ■■■ configuration_compliance/
■■■ archive/
```

No top-level repository modifications are permitted.

22. Security Architecture

The AQELYN Configuration Compliance & Drift Intelligence Engine is the trusted subsystem responsible for continuously validating configuration integrity, enforcing approved baselines, detecting configuration drift, and governing enterprise configuration compliance.

Every configuration assessment shall be:

```
Explainable
Evidence-backed
Policy-governed
Baseline-aware
Risk-aware
Mission-aware
Fully auditable
Continuously reassessed
```

22.1 Security Principles

```
Zero Trust
Defense in Depth
Least Privilege
Continuous Validation
Secure by Design
Policy Enforcement
Explainable Compliance Intelligence
Continuous Monitoring
```

22.2 Authorization Model

Supported operational roles:

```
Configuration Administrator
Compliance Officer
Security Administrator
SOC Analyst
Cloud Administrator
Infrastructure Administrator
Mission Owner
Executive Reviewer
```

All configuration changes and compliance decisions shall be governed through the AQELYN Policy Engine.

22.3 Configuration Integrity

Configuration assessments shall maintain:

```
Unique assessment identifier
Configuration identifier
Asset references
Baseline references
Policy references
Risk score
Confidence indicators
Remediation guidance
Audit trail
```

Assessment history shall be append-only.

22.4 Baseline Integrity

Baseline definitions shall maintain:

```
Baseline identifier
Version
Owner
Approval state
Effective date
Retirement date
Policy mappings
Audit record
```

No baseline shall be considered authoritative without approval state, version metadata, and source lineage.

23. Configuration Lifecycle

23.1 Compliance Lifecycle

```
Configuration Collected
↓
Baseline Comparison
↓
Compliance Assessment
↓
Drift Detection
↓
Remediation Recommendation
↓
Continuous Validation
```

23.2 Baseline Lifecycle

```
Baseline Created
↓
Approved
↓
Published
↓
Applied
↓
Updated
↓
Retired
```

23.3 Audit Lifecycle

```
Assessment Created
↓
Evidence Linked
↓
Policy Validated
↓
Audit Stored
```

24. Continuous Configuration Operations

The engine continuously evaluates:

```
Configuration changes
Baseline deviations
Policy violations
Cloud configuration
Container configuration
Infrastructure changes
Identity configuration
Compliance posture
```

25. Performance Requirements

The engine shall support:

```
Continuous assessment
Low-latency drift detection
Enterprise-scale configuration analysis
Concurrent compliance validation
High availability
Continuous operation
```

26. Scalability Requirements

The engine shall scale to support:

```
Global enterprises
Hybrid infrastructure
Multi-cloud environments
Millions of configuration records
Distributed validation
Long-term configuration history
```

27. Audit Requirements

Every configuration operation shall generate immutable audit records.

Audit events include:

```
Configuration assessed
Configuration changed
Baseline updated
Drift detected
Compliance status changed
Remediation completed
```

28. Failure Handling

28.1 Assessment Failure

```
Assessment retried
Failure recorded
Administrator notified
```

28.2 Drift Detection Failure

```
Detection retried
Previous assessment retained
Audit generated
```

28.3 Baseline Validation Failure

```
Validation repeated
Manual review initiated
Audit recorded
```

28.4 Policy Failure

```
Configuration approval blocked
Policy violation recorded
Manual approval required
```

29. Testing Strategy

29.1 Unit Testing

Validate:

```
Baseline Management Engine
Compliance Assessment Engine
Drift Detection Engine
Configuration Intelligence Engine
Remediation Recommendation Engine
Policy Correlation Engine
```

29.2 Integration Testing

Verify interaction with:

```
Kernel
Knowledge Graph
Security Data Lake
Asset Inventory Engine
Policy Engine
Compliance Engine
Risk Intelligence Engine
AI Decision Engine
Executive Reporting Engine
```

29.3 System Testing

Validate:

```
Baseline validation
Compliance assessment
Drift detection
Configuration intelligence
Remediation recommendations
Audit generation
```

29.4 Security Testing

Verify:

```
Authorization
Policy enforcement
Explainability
Audit logging
Configuration integrity
```

29.5 Regression Testing

Verify IS-001 through IS-025 remain unaffected.

30. Acceptance Criteria

IS-026 is complete when:

```
Baseline Management Engine implemented
Compliance Assessment Engine implemented
Drift Detection Engine implemented
Configuration Intelligence Engine implemented
Remediation Recommendation Engine implemented
Repository unchanged
Testing documented
```

31. Repository Validation

Repository structure remains unchanged.

```
AQELYN/
■■■ src/configuration_compliance/
■■■ tests/configuration_compliance/
■■■ docs/configuration_compliance/
■■■ api/configuration_compliance/
■■■ archive/
```

No top-level repository modifications are permitted.

32. Engineering Summary

IS-026 introduces the AQELYN Configuration Compliance & Drift Intelligence Engine, providing enterprise-scale configuration compliance assessment, baseline governance, drift detection, policy correlation, remediation guidance, configuration intelligence, and explainable compliance reporting.

Major capabilities include:

```
Configuration Baseline Management
Continuous Compliance Assessment
Configuration Drift Detection
Configuration Intelligence
Policy Correlation
Remediation Recommendations
Mission-Aware Compliance Reporting
Executive Compliance Reporting
Continuous Validation
Configuration Governance
```

The engine integrates with all previously completed AQELYN engines while preserving repository stability, modularity, and backward compatibility.

33. Specification Status

```
Specification ID : IS-026
Title : AQELYN Configuration Compliance & Drift Intelligence Engine
```

```
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0026
```

34. EA-0026 Engineering Objective

The objective of IS-026 was to introduce a dedicated Configuration Compliance & Drift Intelligence Engine that enables AQELYN to continuously validate configurations, detect drift, enforce baselines, correlate policy obligations, generate remediation recommendations, and preserve configuration audit history.

The engine extends AQELYN from asset inventory into configuration governance and configuration integrity.

35. EA-0026 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

```
Configuration baseline management
Continuous compliance validation
Configuration drift detection
Configuration intelligence
Policy correlation
Remediation recommendations
Baseline lifecycle governance
Knowledge Graph integration
Security Data Lake integration
Risk and asset integration
AI Decision integration
Executive reporting integration
Event publishing
```

36. Major Engineering Decisions

36.1 Decision 1 - Dedicated Configuration Compliance Engine

Configuration compliance and drift intelligence are implemented as a standalone engine rather than embedded in asset inventory, vulnerability intelligence, or compliance governance.

Rationale:

```
Clear separation of configuration state from asset identity.
Independent lifecycle and governance.
Better support for baseline versioning, drift detection, and configuration integrity.
Improved traceability for configuration compliance decisions.
```

36.2 Decision 2 - Baselines Are First-Class Governance Objects

Baselines are versioned, approved, published, applied, updated, and retired as governed objects.

Benefits:

```
Configuration expectations become explicit.
Compliance assessment can be repeated and audited.
Drift analysis has an authoritative comparison point.
```

36.3 Decision 3 - Drift Detection Is Continuous

Drift is detected continuously rather than only during scheduled audits.

Benefits:

```
Unauthorized or risky changes are detected faster.
Configuration risk becomes visible near real time.
Remediation can be prioritized before incidents occur.
```

36.4 Decision 4 - Configuration Assessments Are Evidence-Backed

Every configuration assessment includes baseline references, policy rationale, evidence, confidence, configuration history, and drift explanation.

Benefits:

```
Compliance decisions become defensible.
Audit trails support regulatory review.
Remediation teams receive explainable findings.
```

36.5 Decision 5 - Event-Driven Configuration Lifecycle

Configuration assessment, drift detection, compliance status, baseline, and remediation events are published through the AQELYN Event Bus.

Examples include:

```
configuration.assessed
configuration.updated
configuration.compliant
configuration.noncompliant
configuration.drift.detected
configuration.drift.resolved
baseline.created
baseline.updated
baseline.retired
remediation.recommended
remediation.completed
```

36.6 Decision 6 - Universal Object Model Extension

New domain objects introduced include:

```
ConfigurationRecord
BaselineDefinition
DriftAssessment
ConfigurationRemediation
```

37. Architectural Integration Summary

Engine	Integration
IS-001 Kernel	Runtime lifecycle and service registration
IS-002 Universal Object Model	Configuration, baseline, drift, remediation objects
IS-003 Event Bus	Configuration, drift, baseline, remediation events
IS-004 Evidence Engine	Evidence references and configuration support
IS-005 Knowledge Graph	Asset, configuration, baseline, policy relationships
IS-006 Trust Engine	Data confidence and configuration trust
IS-007 Mission Engine	Mission-aware compliance prioritization
IS-008 Workflow Engine	Baseline approval and remediation workflows

IS-009 Policy Engine	Configuration governance and policy validation
IS-010 Compliance Engine	Regulatory reporting and control mapping
IS-013 Risk Intelligence Engine	Risk scoring and business impact
IS-019 Security Data Lake	Configuration telemetry and historical configurations
IS-020 AI Decision Engine	Remediation recommendations and learning insights
IS-022 Executive Reporting Engine	Executive compliance summaries
IS-024 Vulnerability Intelligence Engine	Vulnerability and remediation context
IS-025 Asset Inventory Engine	Asset inventory and ownership context

No existing engine required redesign.

38. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/
■■■ src/configuration_compliance/
■■■ tests/configuration_compliance/
■■■ api/configuration_compliance/
■■■ docs/configuration_compliance/
■■■ archive/
```

No top-level directories were added, removed, or renamed.

39. Security Impact Summary

The specification introduces configuration-compliance-specific security controls:

```
Policy-governed configuration changes
Evidence-backed assessments
Baseline version governance
Continuous drift detection
Configuration compliance scoring
Remediation ownership tracking
Assessment audit trail
Compliance mapping
Role-authorized configuration administration
```

No reduction in the security posture of existing components was identified.

40. Capabilities Added

The engine enables AQELYN to support:

```
Configuration baseline management
Continuous compliance validation
Configuration drift detection
Policy compliance assessment
Remediation recommendations
Configuration intelligence
Mission-aware compliance reporting
Executive compliance summaries
Continuous reassessment
Configuration governance
```

41. Risks Identified

Risk	Mitigation
Unauthorized configuration drift	Continuous drift detection and policy enforcement
Baseline ambiguity	Versioned BaselineDefinition objects
Stale configuration data	Continuous assessment and Security Data Lake history
Compliance gaps	Policy Correlation Engine and Compliance Engine integration
Remediation overload	Risk-aware recommendations and prioritization
Poor auditability	Evidence-backed assessments and immutable audit trail
Invalid baseline updates	Approval workflow and version control
Weak configuration source lineage	ConfigurationRecord source and evidence references

No critical architectural risks were identified that require redesign.

42. Verification Summary

The specification defines verification for:

```
Unit testing
Integration testing
System testing
Security testing
Regression testing
```

Acceptance criteria cover baseline management, compliance assessment, drift detection, configuration intelligence, remediation recommendation, repository validation, and testing documentation.

43. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

```
Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline
```

44. Dependencies

Required:

```
EA-0001 through EA-0025
IS-001 through IS-026
```

Enables:

```
IS-027 and subsequent identity threat detection, cloud posture, SaaS posture, supply chain intelligence,
and cyber resilience components
```

45. Completion Record

```
Engineering Archive : EA-0026
Implementation Specification : IS-026
Title : AQELYN Configuration Compliance & Drift Intelligence Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
```

46. Archive Index Update

```
EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine
EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine
EA-0015 IS-015 AQELYN Security Operations Engine
EA-0016 IS-016 AQELYN Digital Forensics Engine
EA-0017 IS-017 AQELYN Threat Detection & Analytics Engine
EA-0018 IS-018 AQELYN Automated Response & Orchestration Engine
EA-0019 IS-019 AQELYN Security Data Lake & Telemetry Platform
EA-0020 IS-020 AQELYN AI Decision Intelligence Engine
EA-0021 IS-021 AQELYN Predictive Analytics & Forecasting Engine
EA-0022 IS-022 AQELYN Executive Intelligence & Strategic Reporting Engine
EA-0023 IS-023 AQELYN Threat Exposure & Attack Surface Management Engine
EA-0024 IS-024 AQELYN Vulnerability Intelligence & Prioritization Engine
EA-0025 IS-025 AQELYN Cyber Asset Discovery & Inventory Intelligence Engine
EA-0026 IS-026 AQELYN Configuration Compliance & Drift Intelligence Engine
```

47. Engineering Phase Status

```
Completed Engineering Archives : EA-0001 through EA-0026

Current Status:
EA-0026 COMPLETE

Next Implementation Specification:
IS-027 - AQELYN Identity Threat Detection & Behavioral Analytics Engine
```

48. Engineering Archive Publication Standard

EA-0026 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/
■
■■■ diagrams/
■ ■■■ Architecture.svg
■ ■■■ Component.svg
■ ■■■ Workflow.svg
■ ■■■ EventFlow.svg
■ ■■■ Integration.svg
■
■■■ examples/
■ ■■■ example_*.md
```

```
■
■■■ HTML/
■ ■■■ index.html
■ ■■■ styles.css
■ ■■■ assets/
■
■■■ images/
■
■■■ journal/
■ ■■■ Engineering_Journal.md
■
■■■ manifest/
■ ■■■ manifest.json
■
■■■ MD/
■ ■■■ EA-xxxx.md
■
■■■ PDF/
■ ■■■ EA-xxxx.pdf
■
■■■ requirements/
■ ■■■ Requirements_Matrix.md
■
■■■ traceability/
■ ■■■ Traceability_Matrix.md
■
■■■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

49. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-026-001	Manage configuration baselines	Sections 8, 12, 23	Complete
FR-026-002	Continuously validate compliance	Sections 8, 12, 23	Complete
FR-026-003	Detect drift	Sections 8, 12, 23	Complete
FR-026-004	Provide explainable compliance intelligence	Sections 8, 22, 36	Complete
FR-026-005	Support governance	Sections 8, 22, 23	Complete
FR-026-006	Publish configuration events	Sections 8, 15, 36	Complete
NFR-026-001	Continuous assessment	Sections 9, 24	Complete
NFR-026-002	Enterprise scalability	Sections 9, 25, 26	Complete
NFR-026-003	Low-latency drift detection	Sections 9, 25	Complete
NFR-026-004	Explainability	Sections 9, 22, 36	Complete
NFR-026-005	Auditability	Sections 9, 27, 39	Complete
NFR-026-006	Repository stability	Sections 21, 31, 38	Complete

50. Traceability Matrix

Source	Target	Relationship
IS-026 Purpose	EA-0026 Objective	Defines why the engine exists
Baseline Management Engine	FR-026-001	Maintains baselines
Compliance Assessment Engine	FR-026-002	Validates compliance
Drift Detection Engine	FR-026-003	Detects deviations

| Configuration Intelligence Engine | Compliance intelligence | Maintains history, health, and risk |

| Remediation Recommendation Engine | Remediation lifecycle | Generates remediation guidance |

| Policy Correlation Engine | Governance and compliance | Maps configurations to policy |

| Event Publisher | FR-026-006 | Publishes configuration events |

| Security Data Lake Integration | Historical configuration data | Supplies telemetry and history |

| AI Decision Integration | Remediation recommendations | Supplies confidence and recommendations |

| Risk & Asset Integration | Asset and risk context | Supplies risk and mission impact |

| Compliance Integration | Audit and control mapping | Supports reporting and evidence |

| Repository Validation | Repository Standard | Confirms no top-level redesign |

51. Engineering Journal

Journal Entry - EA-0026

EA-0026 was created to archive completion of IS-026 - AQELYN Configuration Compliance & Drift Intelligence Engine.

The archive records the expansion of AQELYN into configuration compliance and drift intelligence. IS-026 defines the structure needed to manage baselines, assess compliance, detect drift, maintain configuration intelligence, map policies, generate remediation recommendations, and publish configuration lifecycle events.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Configuration compliance must be modeled separately from asset inventory and vulnerability intelligence. Asset inventory identifies what exists, vulnerability intelligence identifies weaknesses, and configuration compliance determines whether configured state matches approved baselines and policies.

Governance Note

EA-0026 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

52. Examples

52.1 Example Configuration Record

```
configuration_id: CFG-0001
asset_id: ASSET-1001
baseline_version: BASELINE-LINUX-V3
compliance_status: noncompliant
confidence: 0.93
```

52.2 Example Baseline Definition

```
baseline_id: BASELINE-LINUX
version: v3
approval_status: approved
effective_date: 2026-07-07
owner: configuration_security_team
```

52.3 Example Drift Assessment

```
assessment_id: DRIFT-2001
drift_type: unauthorized_service_enabled
confidence: 0.89
detected_at: 2026-07-07T12:00:00Z
```

52.4 Example Configuration Event

```
{
"event_type": "configuration.drift.detected",
"configuration_id": "CFG-0001",
"asset_id": "ASSET-1001",
"baseline_version": "BASELINE-LINUX-V3",
"source_engine": "aqelyn_configuration_compliance_drift_intelligence_engine"
}
```

53. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0026.md
PDF/EA-0026.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_configuration_compliance.md
```

54. Final Archive Statement

EA-0026 is the Engineering Archive for IS-026 - AQELYN Configuration Compliance & Drift Intelligence Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0026 COMPLETE
IS-026 COMPLETE
NEXT: IS-027
```